Q1. DIRECTIONS *for question 1:* Select the correct alternative from the given choices.
Find the remainder when 4^{84} is divided by 6.

- a) **4** ✓ **Your answer is correct**
- b) **2**
- c) **3**
- d) **0**

Q2. DIRECTIONS *for question 2:* Type in your answer in the input box provided below the question.
The ages of Aditi and her mother were both two-digit numbers comprising the same pair of digits. If fifteen years ago, Aditi's age was one-third the age of her mother, find the sum of their present ages (in years).

Q3. DIRECTIONS *for questions 3 to 9:* Select the correct alternative from the given choices.

If $-2 < a < 1$, $-3 < b < -\frac{1}{3}$, $-3 < c < -\frac{1}{3}$ and $d = \frac{ab}{c}$, which of the following is necessarily true?

○ a) $-\frac{2}{9} < d < 18$

○ b) $-18 < d < \frac{2}{9}$

○ c) **$-18 < d < 9$**

○ d) **$-18 < d < 1$**

Q4. DIRECTIONS *for questions 3 to 9:* Select the correct alternative from the given choices.

From the top of a tower, the angles of depression of the top and bottom of a building are observed to be 30° and 60° respectively. If the building is 40 m away from the tower, then what is the height of the building?

a) $40\sqrt{30}$ m

b) $\frac{40}{\sqrt{3}}$ m

c) **60 m**

d) $\frac{80}{\sqrt{3}}$ m ✓Your answer is correct

Q5. **DIRECTIONS** *for questions 3 to 9:* Select the correct alternative from the given choices.
If a trader sells two articles at the same price, one at a loss of 30% and the other at a profit of 20%, then the trader makes approximately

- a) **11.58% profit.**
- b) **11.58% loss.** ✓Your answer is correct
- c) **6% loss.**
- d) **10% loss.**

Q6. DIRECTIONS *for questions 3 to 9:* Select the correct alternative from the given choices. Which of the following is not a perfect square?

- a) $(171)_8$
- b) $(14,641)_8$
- c) $(61)_8$
- d) $(57)_8$

Q7. DIRECTIONS *for questions 3 to 9:* Select the correct alternative from the given choices.

There are n stones, each weighing 1 kg, 2 kg, … n kg respectively. These stones are placed in three bags – A, B and C – in the following manner. Bag A contains the stones weighing 1 kg, 4 kg, 7 kg …; Bag B contain the stones weighing 2 kg, 5 kg, 8 kg….., while Bag C contains the rest of the stones. After this, the Average Weight (AW), defined as $\left(\frac{\text{Total Weight of stones in a bag}}{\text{Number of stones in that bag}}\right)$, is calculated for each bag. It is then found that the AWs of exactly two of the three bags are same as the weight of one of the stones in the respective bags, while the AW of the other bag is different from the weight of any stone in it. Which of the following can be the value of n?

- a) **82**
- b) **99**
- c) **90**
- d) **113**

Q8. DIRECTIONS *for questions 3 to 9:* Select the correct alternative from the given choices.

A triangle ABC is such that its longest side has a length of 10 cm and one of its sides has a length of 5 cm. What is the length of its third side, if the area of the triangle is 20 sq.cm?

a) $\sqrt{65}$ cm

b) $\sqrt{60}$ cm

c) $\sqrt{260}$ cm

d) $\sqrt{175}$ cm

Q9. DIRECTIONS *for questions 3 to 9:* Select the correct alternative from the given choices.

A three-digit number when reversed becomes three-eighths of the original number. How many such three-digit numbers are there?

- a) **0**
- b) **1**
- c) **2**
- d) **More than 2**

Q10. DIRECTIONS *for question 10:* Type in your answer in the input box provided below the question.

The average age of a class of 59 students and their teacher is 22 years and 4 months. If the average age of the class falls by four months when the teacher is excluded, what is the age (in years) of the teacher?

Q11. DIRECTIONS *for question 11:* Select the correct alternative from the given choices.

If x is a real number such that $g(x) = \min(7 - 3x, 3 + 5x)$, then the maximum possible value of $g(x)$ is

a) $\frac{5}{2}$.

b) $\frac{7}{2}$.

c) $\frac{11}{2}$.

d) $\frac{13}{2}$.

Q12. **DIRECTIONS** *for questions 12 to 16:* Type in your answer in the input box provided below the question.
In how many ways can a person pay an amount ₹120 using two-rupee or five-rupee coins?

Q13. DIRECTIONS *for questions 12 to 16:* Type in your answer in the input box provided below the question.

A person, A, starts descending from the first floor of a building to the ground floor, on a descending escalator. Simultaneously, another person, B, starts ascending from the ground floor of the building to the first floor, using the same escalator. If the speed of B is twice that of A, and A and B take 30 steps and 120 steps respectively to reach their destinations, find the number of steps that are visible on the escalator when it is stationary.

Q14. DIRECTIONS *for questions 12 to 16:* Type in your answer in the input box provided below the question.

If $f(x) = x^3 - x^2 - f(x - 1)$, for $x \geq 2$ and $f(1) = 1$, then find the value of $f(25)$.

Q15. DIRECTIONS *for questions 12 to 16:* Type in your answer in the input box provided below the question.

If x, y, z are positive integers, such that $xy + x + y = 215$, $yz + y + z = 161$ and $zx + z + x = 107$, then $(y - z)/(x - z) =$

Q16. DIRECTIONS *for questions 12 to 16:* Type in your answer in the input box provided below the question.

If $p : q = q : r = r : s = 6$, then $\frac{pq + qr + rs}{q^2 + r^2 + s^2} =$

Q17. DIRECTIONS *for questions 17 to 20:* Select the correct alternative from the given choices.
At how many distinct points do any two of the three graphs $x^2 + y^2 = 25$, $4x + 3y = 25$ and $x = 0$ meet each other?

- a) **2**
- b) **3**
- c) **5**
- d) **4**

Q18. **DIRECTIONS** *for questions 17 to 20:* Select the correct alternative from the given choices.

Rakesh purchased a pen and a book for ₹53. The next day, he returned the book and got a full refund, with which he purchased 6 pencils and also got back a change of ₹21. If the price of a pen is twice that of a pencil, then the number of pens that he can purchase with the change that he got is at most

- a) **0**
- b) **1**
- c) **2** ✔ **Your answer is correct**
- d) **3**

Q19. **DIRECTIONS** *for questions 17 to 20:* Select the correct alternative from the given choices.

Find the range of y, if $|2y - 3| \leq 11$.

a) **[−4, 7]**

b) **[3, 7]**

c) **[−2, 7]**

d) **[0, 8]**

Q20. DIRECTIONS *for questions 17 to 20:* Select the correct alternative from the given choices.
If fresh grapes contain 80% water and 20% pulp, by weight, and 10 kg of fresh grapes yield 2.5 kg of dry grapes, then find the percentage of pulp, by weight, in dry grapes.

- a) **20%**
- b) **40%**
- c) **75%**
- d) **80%**

Q21. DIRECTIONS *for question 21:* Type in your answer in the input box provided below the question.

Twenty men were employed to do some work in a certain time. It was found that, after one-third of the scheduled time, only one-quarter of the total work was completed. How many more men should now be employed to complete the work in $3/4^{th}$ of the originally scheduled time?

Q22. DIRECTIONS *for question 22:* Select the correct alternative from the given choices.

There are some baskets labelled 1, 2, 3, 4 …… and so on upto $2n$, where $n > 6$, such that, for $k = 1, 2, 3, 4, \ldots 2n$, there are exactly k baskets labelled k.

Now, consider the following two cases for the number of apples a_i in the basket labelled i.

Case I : $a_i = i + 1$, if i is odd.
$= i + 2$, if i is even.
Case II : $a_i = i + 2$, if i is odd.
$= i + 1$, if i is even.

If the total number of baskets is N, then find the difference between the total number of apples in the N baskets in Case I and Case II.

a) **1**

b) $\frac{\sqrt{8N+1}-1}{4}$

c) $\frac{N(N-1)}{6}$

d) $\frac{N-4}{8}$